# Matrix and Vector Notation

1. Given $\begin{bmatrix} 2 & \frac{1}{2} \\ -3 & 4 \end{bmatrix}$ and $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, calculate the matrix-vector product.

## Display notation

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \end{bmatrix}$$

$$\begin{pmatrix} \frac{1}{3} \\ x^2 \\ 2+3i \end{pmatrix}$$

$$\begin{vmatrix} 2 & -1 \\ 5 & 3 \end{vmatrix}$$

# Solutions

1. The result is $\begin{pmatrix} 1 \\ -11 \end{pmatrix}$.